MACS HAART Response at Year 2: Frequentist and Bayesian Linear Regression by Baseline Hard Drug Use

Emily Kokesh

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1 Introduction

1.1 Project Description and Data

This project uses data from the Multicenter AIDS Cohort Study (MACS), a long-running cohort of HIV-1 infected men. After initiation of highly active antiretroviral therapy (HAART), MACS collects annual laboratory markers and self-reported health outcomes; this report evaluates treatment response two years after starting HAART.

Year 0 represents each participant's last untreated visit immediately before HAART initiation, and Years 1–8 represent follow-up visits on treatment. This analysis focuses on Year 2 outcomes and uses baseline (Year 0) hard drug use as the main exposure, defined using self-reported use of hard drugs at Year 0 and treated as a fixed baseline characteristic. Treatment response is measured using four Year 2 outcomes: viral load (VLOAD; HIV copies/mL), CD4+ T-cell count (LEU3N; cells/mm^3), and aggregate physical and mental quality of life scores from the SF-36 (AGG_PHYS and AGG_MENT; 0–100 with higher scores indicating better quality of life). Untreated HIV infection is generally characterized by increasing viral load and declining CD4 count; after HAART initiation, viral load is expected to decrease and CD4 count to recover, while quality-of-life changes are less predictable due to competing effects of improved health and treatment burden.

1.2 Research Questions and Hypotheses

This report addressed two questions: whether baseline hard drug use (Yes vs. No) is associated with Year 2 treatment response across four outcomes (VLOAD, LEU3N, AGG_PHYS, AGG_MENT),

and whether adding Year 2 adherence changes these associations. We hypothesized poorer biologic response among baseline hard drug users (higher viral load, lower CD4 count) and potentially lower QoL, and we report parallel frequentist and Bayesian linear regression results using 95% confidence intervals and p-values (frequentist) and posterior means, 95% HDIs, and posterior probabilities (Bayesian).

2 Methods

2.1 Data Management, Cleaning, and Transformations

Baseline hard drug use was defined at Year 0 and treated as a fixed exposure for Year 2 comparisons; demographic and behavioral covariates were recoded into binary analytic categories (White [NH] vs. Other, College Degree vs. No College Degree, Current vs. Non-current Smoker) and Year 2 medication adherence was analyzed as $\geq$95% vs. <95%. BMI values exceeding 80 kg/m^2 or coded as -1 or 999 were recoded as NA (74 records across 52 participants), and outcomes/covariates were screened for implausible values; no negative viral load values were identified and 15 observations exceeding 1,000,000 copies/mL were flagged but retained per investigator instructions.

Viral load (VLOAD) was modeled on the $\log_{10}$ scale due to heavy right-skewness, and the baseline hard drug use effect is reported as a fold-change (10^{β}). CD4+ T-cell count (LEU3N) and the SF-36 aggregate physical and mental quality-of-life scores (AGG_PHYS, AGG_MENT) were modeled on their original scales.

The complete-case analytic cohort used in all models included 463 participants (427 non-users, 36 baseline hard drug users). Cohort flow and exclusions are summarized in the Results.

2.2 Statistical Analysis

For each Year 2 outcome, we fit two adjusted models: (1) a primary model excluding adherence and (2) a secondary model adding Year 2 adherence. Covariates included in adjustment were pre-specified based on investigator guidance, and therefore all standard baseline covariates provided for the project were retained; all analyses used one observation per participant at Year 2 and adjusted for the baseline (Year 0) value of the outcome and baseline covariates, summarizing how the baseline hard drug use estimate changes after adding adherence.

2.3 Frequentist Linear Regression

Multivariable linear regression models were fit separately for each Year 2 outcome: $\log_{10}$(VLOAD), LEU3N, AGG_PHYS, and AGG_MENT. The primary predictor was baseline hard drug use (Yes vs. No). For each outcome, we fit (1) a primary model adjusted for the corresponding Year 0 outcome value and baseline covariates (age, BMI, race, education, smoking) and (2) an adherence-adjusted model adding Year 2 adherence ($\geq$95% vs. <95%). Results are reported as regression coefficients with 95% confidence intervals and two-sided p-values based on Wald t-tests of the hard drug use coefficient using the residual degrees of freedom ($\alpha = 0.05$). Viral load effects are additionally reported as fold-changes (10^{β}). To evaluate whether adherence meaningfully influenced the estimated association, we summarized the change in the baseline hard drug use estimate after adding adherence, reported as the percent change from the primary model to the adherence-adjusted model.

Model assumptions were evaluated using residuals versus fitted plots, residual histograms, and normal Q–Q plots; residuals were generally centered around zero without strong curvature, with mild heteroscedasticity and modest tail departures for LEU3N and quality-of-life outcomes that were similar in models with and without adherence.

2.3.1 Bayesian Regression

Bayesian linear regression models were fit for each Year 2 outcome using the same covariates as the frequentist analyses and were implemented in Stan. Weakly informative priors were used: regression coefficients $\beta_j \sim \text{Normal}(0, 1000^2)$ and residual standard deviation $\sigma \sim \text{Half-Normal}(0, 100^2)$. Posterior sampling used 4 chains with 1,000 warmup iterations and 10,000 sampling iterations per chain.

For each outcome, we fit models without and with adherence and summarized the baseline hard drug use coefficient using the posterior mean and 95% highest posterior density interval (HPDI). Evidence of a difference between baseline hard drug use groups was assessed using the posterior distribution of the hard drug use effect. Effects were considered credibly different from the null when the 95% HDI excluded 0 (for β) or excluded 1 after transformation to fold-change (for 10^{β}). We also report posterior probabilities for the direction of the effect ($\Pr(10^{\beta} > 1)$ for viral load and $\Pr(\beta < 0)$ for outcomes where lower values indicate worse response).

Model convergence was assessed using $\hat{R}$ and effective sample size (ESS), along with trace plots, autocorrelation plots, and density overlays across chains for key parameters (intercept, baseline hard drug use coefficient, and σ).

2.3.2 Comparison strategy: frequentist vs. Bayesian inference

Frequentist and Bayesian results were compared for the same target parameter: the adjusted association between baseline hard drug use (Yes vs. No) and each Year 2 outcome. Comparisons emphasized the direction and magnitude of estimated effects and agreement between 95% confidence intervals and 95% HPDIs, as well as qualitative conclusions from frequentist p-values and Bayesian posterior probabilities about whether the hard drug use effect was statistically different from 0 and its direction. The impact of adherence adjustment was evaluated equivalently in both methods by comparing the hard drug use effect estimate percent change from models fit without versus with adherence.

3 Results

3.1 Study Overview and Analytic Sample

The dataset included 715 unique participants with a Year 0 (pre-HAART) visit, and 506 also had a Year 2 visit. The complete-case analytic cohort included 463 participants, of whom 36 (7.8%) reported baseline hard drug use and 427 (92.2%) did not. The primary reason for exclusion was absence of a Year 2 visit ($n = 209$). Additional exclusions were due to missing outcomes or baseline covariates.

3.2 Baseline Characteristics (Year 0)

Table 1: Baseline characteristics of participants by baseline hard drug use status

Variable	**Overall (N = 463)**	**No (N = 427)**	**Yes (N = 36)**	***p*-value**
Age (years)	43 (9)	43 (9)	44 (10)	0.633
BMI (kg/m^2)	25.2 (4.3)	25.3 (4.4)	23.6 (3.4)	0.007
Race/Ethnicity				0.204
White (NH)	296 (64%)	277 (65%)	19 (53%)	
Other	167 (36%)	150 (35%)	17 (47%)	
Education level				0.073
No college degree	262 (57%)	236 (55%)	26 (72%)	
College+	201 (43%)	191 (45%)	10 (28%)	
Smoking status				< 0.001
Non-current smoker	287 (62%)	278 (65%)	9 (25%)	
Current smoker	176 (38%)	149 (35%)	27 (75%)	
Baseline viral load	32,590 (10,812, 137,021)	33,001 (10,775, 137,016)	29,256 (12,907, 293,875)	0.879
Baseline CD4+ count	373 (202)	374 (202)	360 (201)	0.681
Baseline physical QoL	51 (9)	51 (9)	49 (7)	0.088
Baseline mental QoL	45 (14)	45 (14)	42 (12)	0.133

Note: Values are mean (SD) for continuous variables, n (%) for categorical variables, and median (Q1, Q3) for viral load.
p-values from Welch two-sample t-tests (continuous), Pearson chi-squared tests (categorical), and Wilcoxon rank-sum test (viral load).

Baseline characteristics are summarized in Table 1 for the analytic cohort (N=463) by baseline hard drug use (No: n=427, Yes: n=36), with p-values from Welch two-sample t-tests (continuous variables), Pearson chi-squared tests (categorical variables), and the Wilcoxon rank-sum test (viral load).

Two baseline differences met the $\alpha = 0.05$ threshold. Baseline hard drug users had lower mean BMI than non-users (23.6 vs. 25.3 kg/m^2; $p = 0.007$) and were more likely to be current smokers (75% vs. 35%; $p < 0.001$). Other baseline characteristics were broadly similar between groups: age (44 vs. 43 years; $p = 0.633$), race/ethnicity ($p = 0.204$), education ($p = 0.073$), baseline viral load (median 29,256 vs. 33,001 copies/mL; $p = 0.879$), baseline CD4 count (360 vs. 374 cells/mm^3; $p = 0.681$), and baseline physical and mental QoL scores ($p = 0.088$ and $p = 0.133$, respectively).

3.3 Year 2 Outcomes and Adherence

Table 2: Treatment response outcomes and adherence at Year 2 by baseline hard drug use.

Variable	**Overall (N = 463)**	**No (N = 427)**	**Yes (N = 36)**	***p*-value**
Year 2 adherence				0.228
At least 95%	417 (90%)	382 (89%)	35 (97%)	
Below 95%	46 (9.9%)	45 (11%)	1 (2.8%)	
Year 2 viral load	31 (10, 275)	31 (10, 287)	44 (9, 161)	0.798
Year 2 $\log_{10}$(VLOAD)	1.79 (1.23)	1.79 (1.21)	1.85 (1.45)	0.807
Year 2 CD4+ count	542 (267)	556 (264)	371 (249)	< 0.001
Year 2 physical QoL	50 (10)	50 (10)	44 (12)	0.009
Year 2 mental QoL	47 (12)	48 (12)	46 (14)	0.492

Note: Values are n (%) for adherence, median (Q1, Q3) for viral load, and mean (SD) for other continuous variables.

p-values from Pearson chi-squared tests (adherence), Wilcoxon rank-sum test (viral load), and Welch two-sample t-tests (other continuous variables).

Year 2 adherence and outcomes are summarized in Table 2. Adherence was high in both groups (97% vs. 89% with $\geq$95% adherence; $p = 0.228$). Year 2 viral load did not differ by baseline hard drug use (median 44 vs. 31 copies/mL, $p = 0.798$), and mean $\log_{10}$(VLOAD) was also similar ($p = 0.807$). In contrast, baseline hard drug users had lower mean Year 2 CD4 counts (371 vs. 556 cells/mm^3, $p < 0.001$) and lower mean physical QoL (44 vs. 50, $p = 0.009$). Mean mental QoL was comparable between groups (46 vs. 48, $p = 0.492$).

3.3.1 Frequentist Regression Results

Table 3: Table 3. Association of baseline hard drug use (Yes vs No) with Year 2 outcomes: frequentist models with and without adherence adjustment.

	Model Without ADH		**Model With ADH**		
Outcome	**Est.** (95% CI)	p	**Est.** (95% CI)	p	**% change** (No→With)
Viral load (fold-change; 10^{β})	0.872 (0.349, 2.180)	0.770	0.971 (0.389, 2.424)	0.950	11.3%
CD4+ T-cell count (cells/mm^3)	−163.880 (−226.345, −101.415)	< 0.001	−169.683 (−232.215, −107.151)	< 0.001	3.5%
Aggregate physical QoL score	−3.336 (−6.062, −0.610)	0.017	−3.567 (−6.298, −0.836)	0.011	6.9%
Aggregate mental QoL score	−0.289 (−3.752, 3.174)	0.870	−0.531 (−4.004, 2.942)	0.764	84.0%

Note: Estimates are for baseline hard drug use (Yes vs No). Viral load is modeled as $\log_{10}$(VLOAD) and reported as fold-change (10^{β}). Other outcomes are reported on their original scale. All models adjust for baseline outcome value and baseline covariates (age, BMI, race, education, smoking). The "% change" column is the percent change in the reported estimate after adding adherence.

Frequentist regression results are shown in Table 3. Without adherence adjustment, baseline hard drug use was not significantly associated with Year 2 viral load or Year 2 mental QoL, but was significantly associated with lower mean Year 2 CD4 count ($\beta = -163.880$ cells/mm^3, 95% CI: -226.345 to -101.415, $p < 0.001$) and lower mean Year 2 physical QoL ($\beta = -3.336$, 95% CI: -6.062 to -0.610, $p = 0.017$). After adding adherence, the mental QoL estimate changed by 84% but remained not statistically significant with a confidence interval spanning 0 ($\beta = -0.531$, 95% CI: -4.004 to 2.942, $p = 0.764$). Adding adherence changed estimates only modestly overall and did not alter conclusions for any outcome.

3.3.2 Bayesian Regression Results

Table 4: Table 4. Association of baseline hard drug use (Yes vs No) with Year 2 outcomes: Bayesian models with and without adherence adjustment.

	Model Without ADH		**Model With ADH**		
Outcome	**Mean** (95% HDI)	**Pr(Effect > 0/1)**	**Mean** (95% HDI)	**Pr(Effect > 0/1)**	**% change** (No→With)
Viral load (fold-change; 10^{β})	0.973 (0.250, 1.908)	0.385	1.085 (0.283, 2.134)	0.476	11.6%
CD4+ T-cell count (cells/mm^3)	−163.664 (−226.157, −102.452)	< 0.001	−169.437 (−231.065, −106.046)	< 0.001	3.5%
Aggregate physical QoL score	−3.345 (−6.086, −0.622)	0.008	−3.555 (−6.260, −0.760)	0.006	6.3%
Aggregate mental QoL score	−0.277 (−3.779, 3.187)	0.440	−0.527 (−3.965, 3.017)	0.380	90.5%

Note: Estimates are posterior means for baseline hard drug use (Yes vs No). 95% HPDI = 95% highest posterior density interval. Viral load is modeled as $\log_{10}$(VLOAD) and reported as fold-change (10^{β}); for viral load, Pr(Effect > 0/1) is Pr(fold-change > 1). For other outcomes, Pr(Effect > 0/1) is Pr($\beta > 0$). All models adjust for baseline outcome value and baseline covariates (age, BMI, race, education, smoking).

Bayesian regression results are shown in Table 4. Results closely matched the frequentist analyses: there was little evidence of an association between baseline hard drug use and mean Year 2 viral load (fold-change mean = 0.973, 95% HPDI: 0.250 to 1.908, Pr(fold-change > 1) = 0.385) or mental QoL (mean $\beta = -0.277$, 95% HPDI: −3.779 to 3.187, Pr($\beta > 0$) = 0.440). In contrast, baseline hard drug use was associated with lower mean Year 2 CD4 count (mean $\beta = -163.664$, 95% HPDI: −226.157 to −102.452) and lower mean physical QoL (mean $\beta = -3.345$, 95% HPDI: −6.086 to −0.622). Although the mental QoL estimate changed by 90.5% after adding adherence, the adherence-adjusted estimate remained non-significant with a HPDI spanning 0 ($\beta = -0.527$, 95% CI: −3.965 to 3.017, $p = 0.380$). Adding adherence produced only modest changes in the hard drug use estimates and did not change conclusions for any outcome.

Bayesian diagnostics. Across all models, convergence and mixing were supported by $\hat{R} < 1.05$ and ESS > 1000 for key parameters. Trace plots showed stable chains with good mixing, density overlays showed close agreement across chains, and autocorrelation decayed quickly within a few lags. These diagnostics were similar across models and supported using posterior draws for inference.

3.3.3 Comparison of Frequentist and Bayesian Inference

Table 5: Association of baseline hard drug use (Yes vs No) with Year 2 outcomes: frequentist and Bayesian results with and without adherence adjustment.

	Model Without ADH			Model With ADH		
Outcome	**Freq. Est.** (95% CI)	p	**Bayes Mean** (95% HPDI)	**Freq. Est.** (95% CI)	p	**Bayes Mean** (95% HPDI)
Viral load (fold-change; 10^{β})	0.872 (0.349, 2.180)	0.770	0.973 (0.250, 1.908)	0.971 (0.389, 2.424)	0.950	1.085 (0.283, 2.134)
CD4+ T-cell count (cells/mm^3)	−163.880 (−226.345, −101.415)	< 0.001	−163.664 (−226.157, −102.452)	−169.683 (−232.215, −107.151)	< 0.001	−169.437 (−231.065, −106.046)
Aggregate physical QoL score	−3.336 (−6.062, −0.610)	0.017	−3.345 (−6.086, −0.622)	−3.567 (−6.298, −0.836)	0.011	−3.555 (−6.260, −0.760)
Aggregate mental QoL score	−0.289 (−3.752, 3.174)	0.870	−0.277 (−3.779, 3.187)	−0.531 (−4.004, 2.942)	0.764	−0.527 (−3.965, 3.017)

Note: Estimates correspond to baseline hard drug use (Yes vs No). Viral load models use $\log_{10}$(VLOAD); effects are reported as fold-change (10^{β}). Other outcomes are reported on their original scales. All models adjust for baseline outcome value and baseline covariates (age, BMI, race, education, smoking). Bayesian results summarize posterior means and 95% highest density intervals (HPDIs).

Frequentist and Bayesian results were consistent across outcomes (Table 5), with similar effect directions, magnitudes, and uncertainty intervals with and without adherence. In both frameworks, baseline hard drug use was significantly associated with lower Year 2 CD4 count and lower physical QoL, while viral load and mental QoL differences were not statistically significant. Adding adherence did not change conclusions in either approach.

4 Discussion and Summary

In this secondary analysis of MACS participants initiating HAART, baseline hard drug use was uncommon (7.8%) but was associated with poorer Year 2 immunologic recovery and physical functioning. After adjustment for baseline outcome values and key baseline covariates (age, BMI, race, education, smoking), baseline hard drug use was associated with lower mean Year 2 CD4 count (about 164–170 cells/mm^3 lower) and lower mean physical QoL (about 3–4 points lower), while viral load (fold-change on the $\log_{10}$ scale) and mental QoL were not significantly different. Frequentist and Bayesian results were closely aligned, and adding Year 2 adherence produced only modest changes in estimates and did not change conclusions, suggesting adherence at Year 2 does not explain the observed associations.

Several limitations should be considered. The analysis used a complete-case cohort and excluded 209 participants without a Year 2 visit, so selection bias is possible if follow-up is related to drug use and outcomes. Baseline hard drug use was self-reported at Year 0 and treated as fixed, and adherence was measured only at Year 2 and dichotomized at 95%, which may not capture changes

in use or adherence over time. Residual confounding is also possible because additional clinical and psychosocial factors were not included. Despite these limitations, results were consistent across frequentist and Bayesian approaches, supporting an association between baseline hard drug use and poorer Year 2 CD4 recovery and physical quality of life in this cohort.

5 Reproducible Research Information

- **GitHub Repository:** `https://github.com/emkokesh/BIOS6624`
- **Raw dataset:** `C:/Users/kokeshem/Spring2026/AdvancedStatisticalMethods/Project1/hiv_6624_final.csv`
- **Final analytic dataset used for modeling:** `C:/Users/kokeshem/Spring2026/AdvancedDataAnalysis/BIOS6624/Project1/DataProcessed/project1_clean_final.csv`
- **Analysis scripts (GitHub):** `Project1_EDA.Rmd`, `Project1_Analysis_Frequentist.Rmd`, `Project1_Analysis_Bayesian.Rmd`